**5.** The table gives information about two kettles.

| Kettle | Power (W) | Time to boil 0.5 litres of water (minutes) | Time to boil 0.5 litres of water (hours) |
|---|---|---|---|
| A | 3000 | 2 | 0.033 |
| B | 2400 | 2.5 | |

(a) (i) Calculate the units used (kWh) by **kettle B** if it is used to boil 0.5 litres of water 6 times a day. [2]

units used = .................................. kWh

(ii) Kettle B is used 6 times a day.
Calculate the cost of using kettle B for **one week**. [2]
Cost of 1 unit = 34 p.

cost = ......................................... p

(iii) James suggests that it would be cheaper to use kettle B because it has a lower power.
Sophia disagrees and suggests it would make no difference.
Explain who is correct. [2]
Space for calculation.

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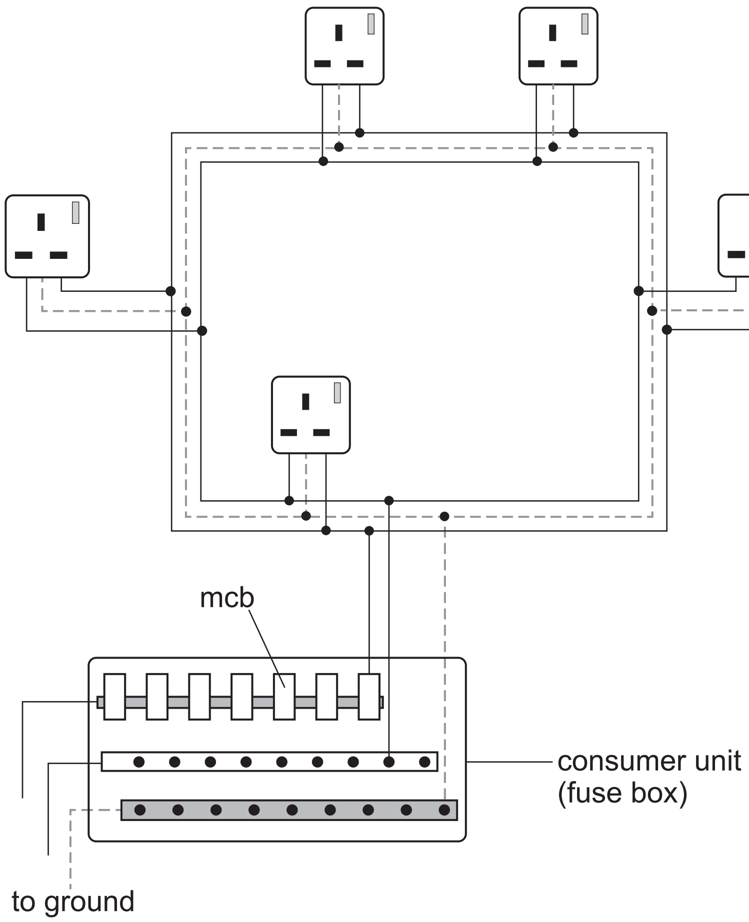
Examiner only

(b) Kettle B is connected into a ring main circuit.
A diagram of a ring main circuit is shown below.




(i) State **two** advantages of a ring main circuit. [2]

1. ..............................................................................................................................

2. ..............................................................................................................................

(ii) The ring main is protected by an mcb.
State **two** advantages of an mcb compared to a fuse. [2]

1. ..............................................................................................................................

2. ..............................................................................................................................



(iii) The mcb has a rating of 32 A.
The maximum power that the ring main can supply is 7360 W.
The table shows some of the appliances that are used in a kitchen.

| Appliance | Power (W) |
|---|---|
| kettle | 2400 |
| toaster | 1200 |
| microwave | 800 |
| washing machine | 750 |
| dishwasher | 1500 |
| iron | 1100 |

Jack states that all these appliances can be connected to the ring main and used at the same time.
Determine whether Jack is correct.
Space for calculation. [1]

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